

REPLACEMENT PARTS MANUAL

HEAT KING HK150

Mobile Glycol Heating System





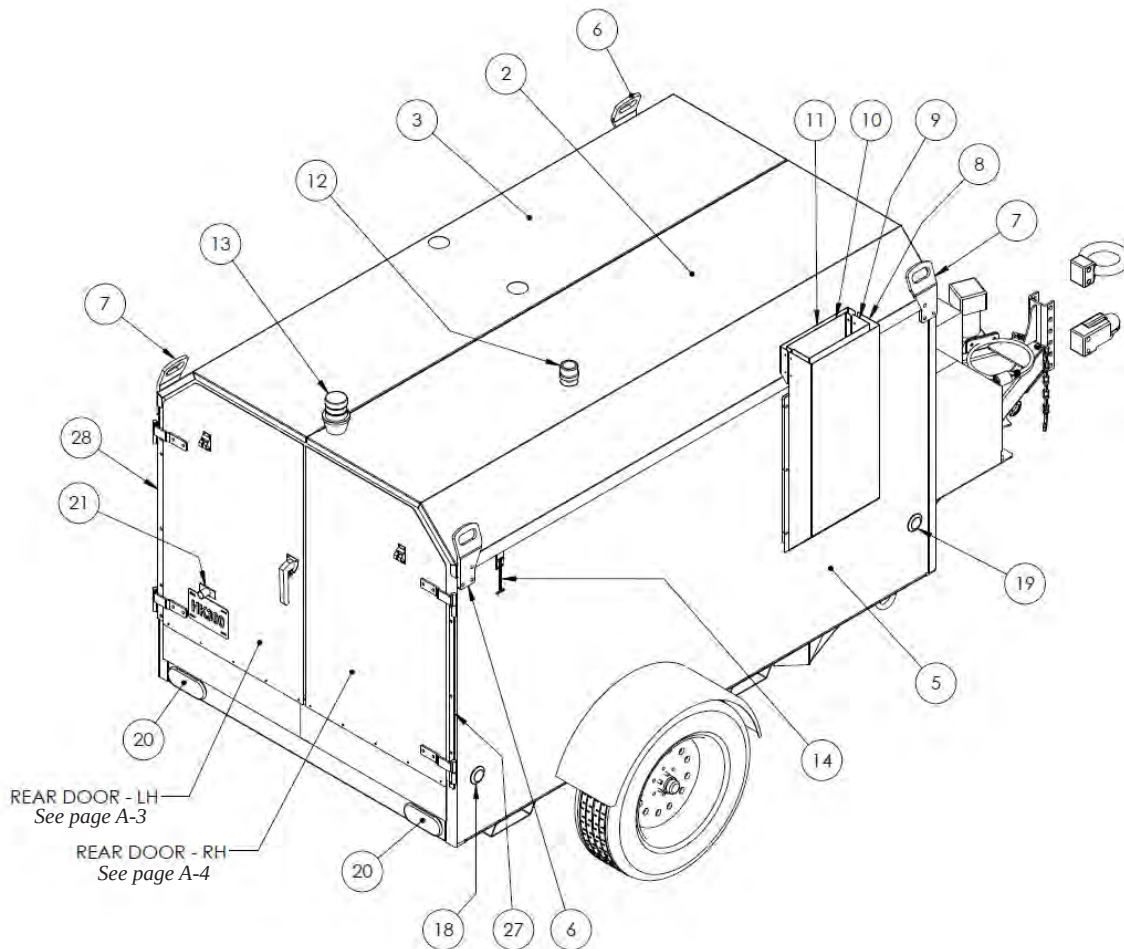
Disclaimer

This manual is proprietary to Heat King Division of Tamarack Industries (hereafter “Heat King”), and may be revised from time to time without notice. Heat King disclaims any understanding to provide you with such revisions, if any. This manual is copyrighted with all rights reserved. No portion of this manual may be copied or reproduced by any means without the prior written consent of Heat King. While every precaution has been taken in the preparation of this document, Heat King assumes no liability to any party for any loss or damage caused by errors or omissions or by statements resulting from negligence, accident, or any other cause. Heat King further assumes no liability arising out of the application or use of any product or system described, herein; nor any liability for incidental or consequential damages arising from the use of this document. Heat King disclaims all warranties of merchantability of fitness for a particular purpose. Heat King reserves the right to make changes without further notice to any product or system described herein to improve reliability, function, or design.

Page Index

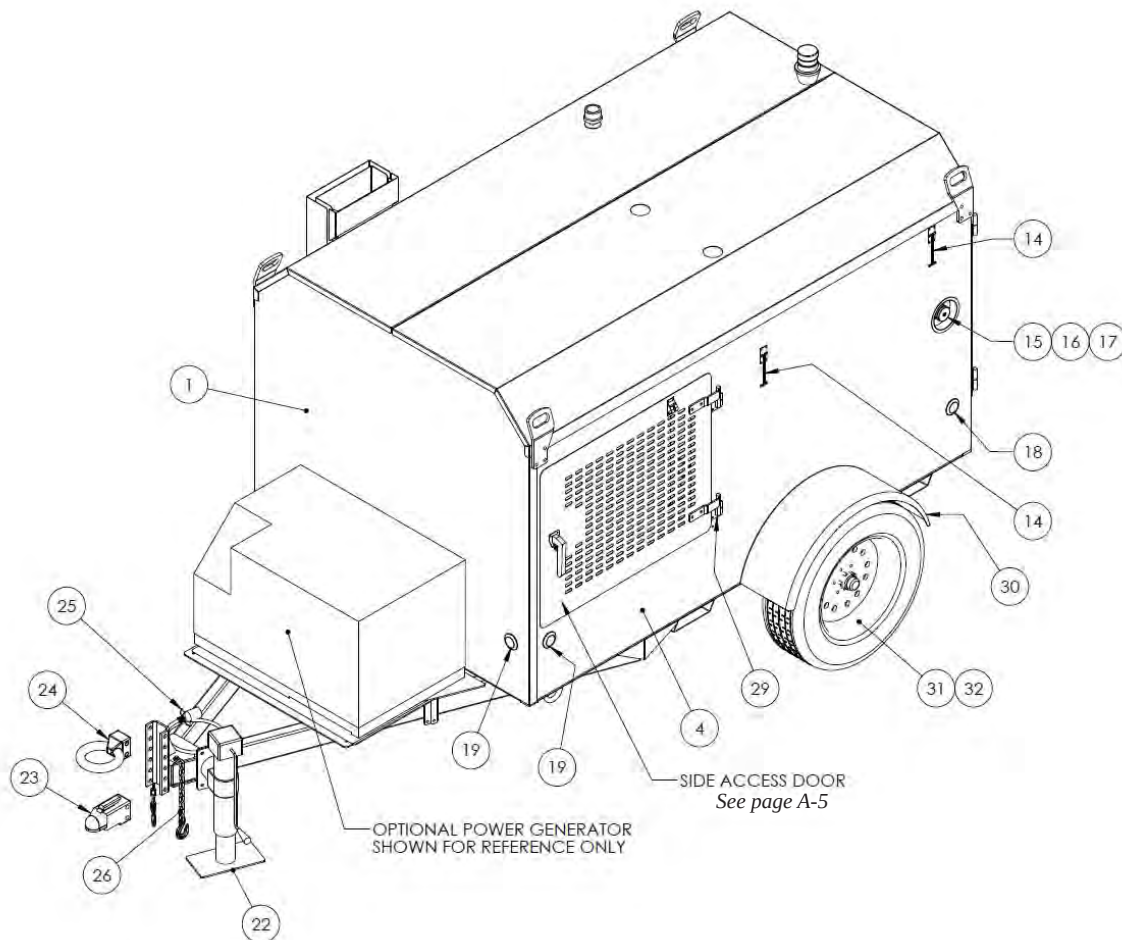
Item Name	Page #
Trailer	Chapter A
Trailer Exterior Hardware	A-1, A-2
Side Access Door	A-3
Rear Door - Left Hand	A-4
Rear Door - Right Hand	A-5
Pump Plate and Heater	Chapter B
Pump Plate Assembly	B-1
Heating Coil Assembly	B-2
Burner Assembly	B-3, B-4
Circulation Pump Assembly,	B-5, B-6
Field Pump Assembly,	B-7
Pump Assembly Breakdown	B-8
Fuel Filter Assembly	B-9
Control Box	Chapter C
Control Box Assembly	C-1
Fuel Tank	Chapter D
Fuel Tank Assembly	D-1
Glycol System	Chapter E
Glycol Tank Assembly	E-1, E-2
Supply Manifold	E-3
Return Manifold	E-4
Reel	Chapter F
Reel Assembly	F-1
Electrical System Schematics	Chapter G
Panel Box Electrical Schematic	G-1
Inside Trailer Electrical Schematic	G-2
Panel Door Electrical Schematic	G-3
Company Contact Information	Back Cover Page

Trailer Exterior Hardware, HK 150



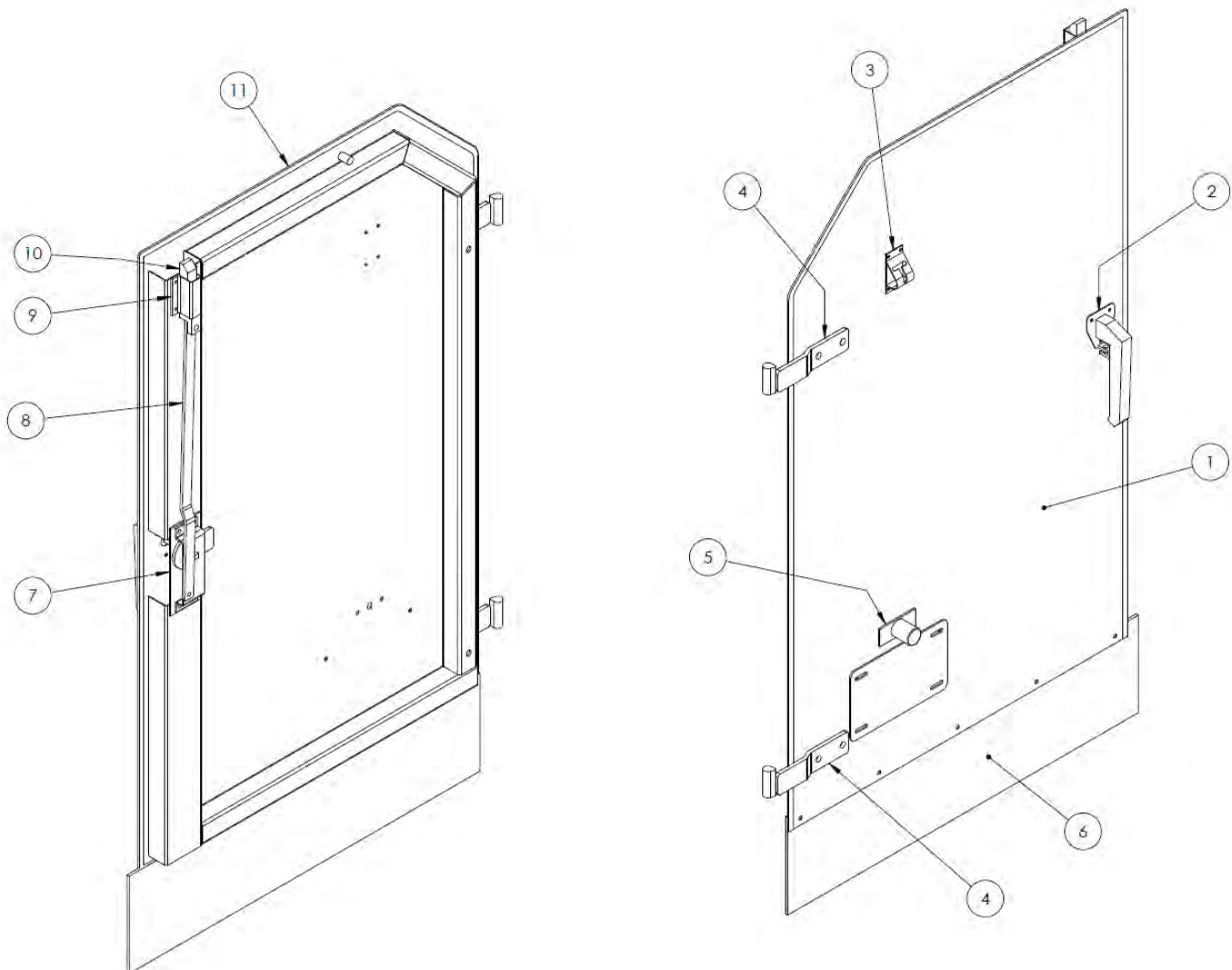
Ref.	Part	Description	Qty.
1	448097	PANEL, FRONT, HK 150	1
2	448170	PANEL, ROOF, HK 150	1
3	448133	PANEL, ROOF, HK 150	1
4	448171	PANEL, SIDE, HK 150	1
5	448134	PANEL, SIDE, HK 150	1
6	448204	HOOK, LIFT, RHS	2
7	448206	HOOK, LIFT, LHS	2
8	451111	EXHAUST, GUARD	1
9	451110	EXHAUST, TERMINAL	1
10	451113	EXHAUST, TERMINAL	1
11	451112	CAP, EXHAUST	1
12	445165	FUEL TANK BREATHER	1
13	446707	STROBE LIGHT	1
14	448009	DOOR KEEPER , 6"	3
15	485134	FILLER NECK	1
16	440232	FLANGE, FILLER NECK	1

Trailer Exterior Hardware HK 150 - Cont'd



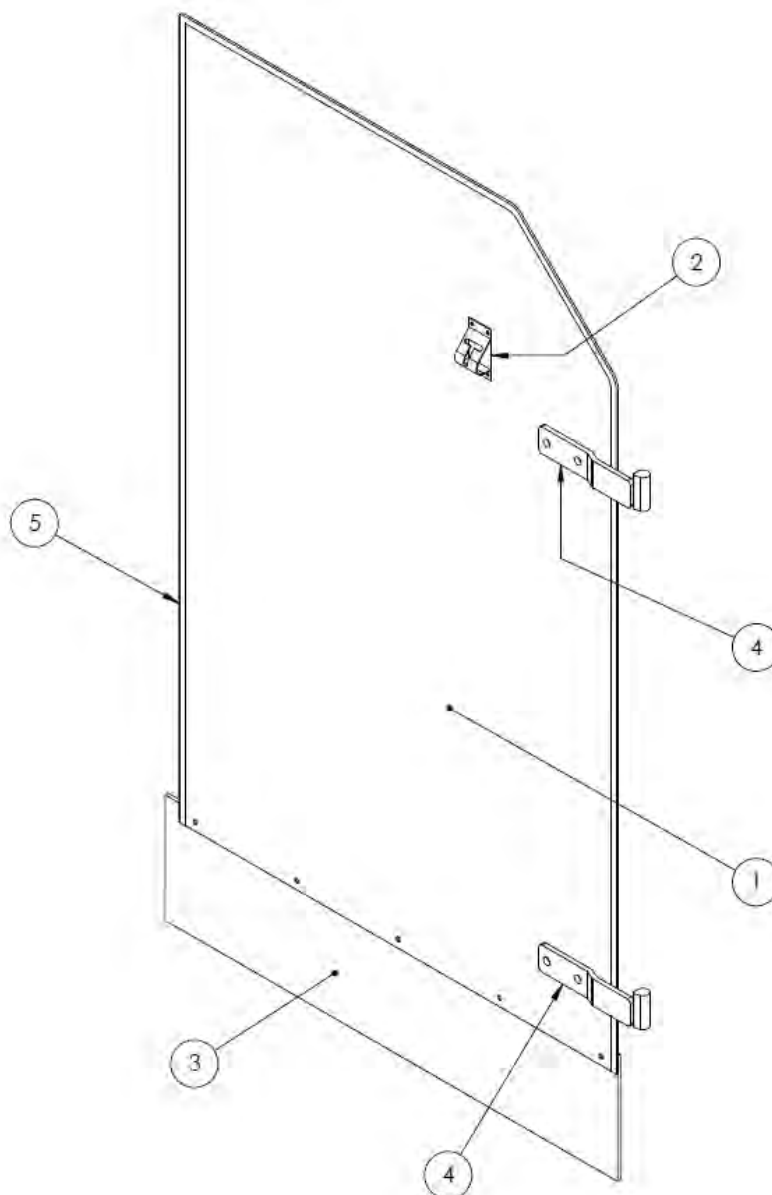
Ref.	Part	Description	Qty.
17	451305	CAP, GAS , LOCKING	1
17*	451301	CAP, GAS , NON-LOCKING	1
18	477008	LAMP, RED, CLEARANCE 2"	2
19	477007	LAMP, ORANGE, CLEARANCE 2"	4
20	477009	LAMP, STOP	2
21	448021	LIGHT, LICENCE PLATE, HK	2
22	448004	TRAILER JACK, 8000 LB	1
23	448003	BALL COUPLER, 2-5/16"	1
24	449025	PINTLE COUPLER	1
25	448017	CABLE, ARTIE RV w/7 Pin CONNECTOR	1
26	448016	CHAIN, SAFETY 3/8" x 30" GR 43	2
27	448213	REAR DOOR HINGE ASSY. - RH	1
28	448212	REAR DOOR HINGE ASSY. - LH	1
29	448191	SIDE DOOR HINGE	2
30	448002	FENDER, 10.75 x 36	2
31	448001	RIM AND TIRE ASSY. (225-75-R15)	2
32	448000	AXLE ASSY.	1

Rear Door Assembly - Left Hand, HK 150



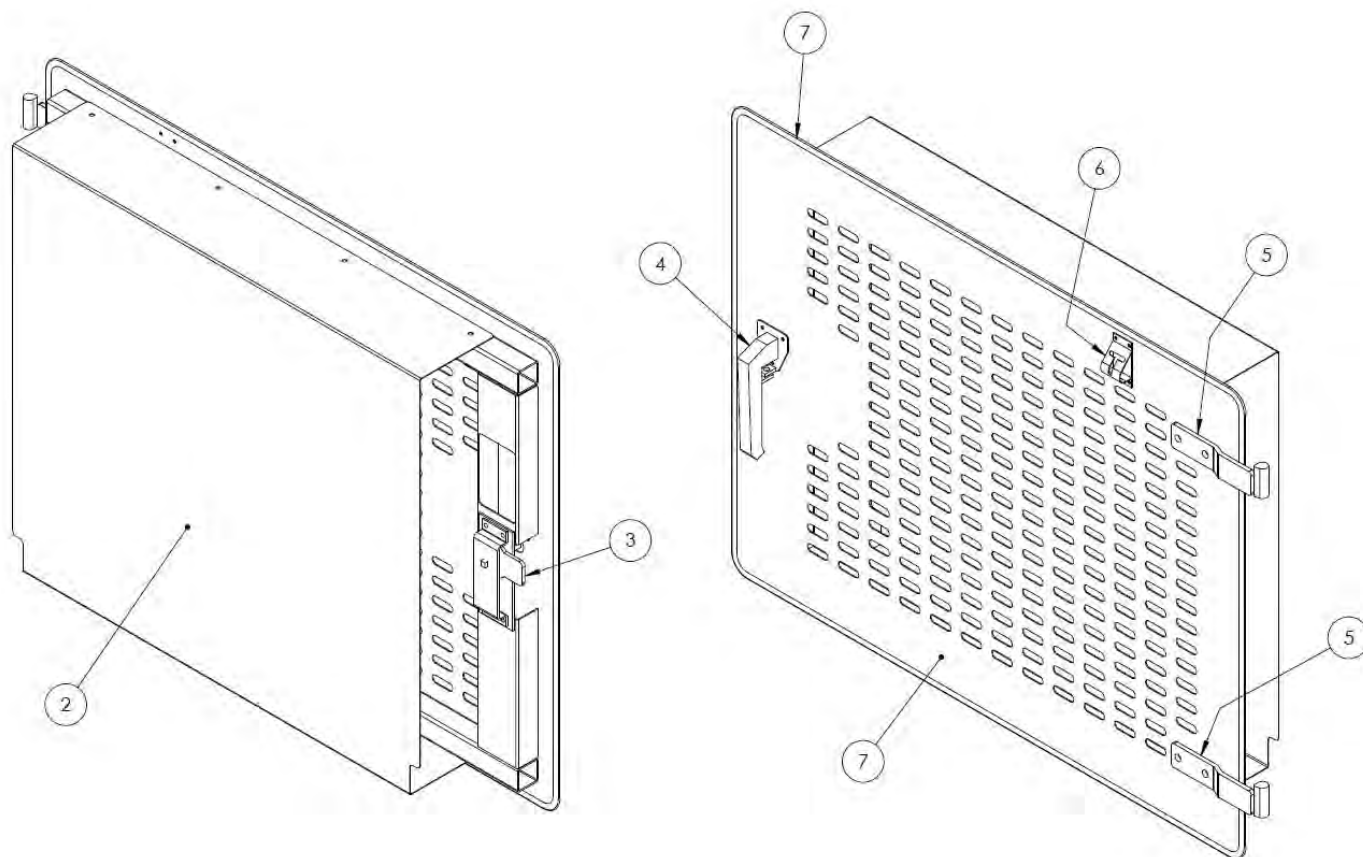
Ref.	Part	Description	Qty.
1	448150	445155 DOOR REAR, LH	1
2	448006	HANDLE, DOOR	1
3	448009	KEEPER, DOOR , 6 IN	1
4	448178	SIDE DOOR HINGE ASSY.	2
5	448021	LIGHT, LICENCE PLATE	1
6	448011	FLAP, 8 IN, REAR DOOR	1
7	448007	LATCH, 3 POINT, CENTER CASE	1
8	448093	ROD	1
9	448091	CASE BOLT	1
10	448092	BOLT	1
11	447160	SEAL, BULB	1

Rear Door Assembly - Right Hand, HK 150



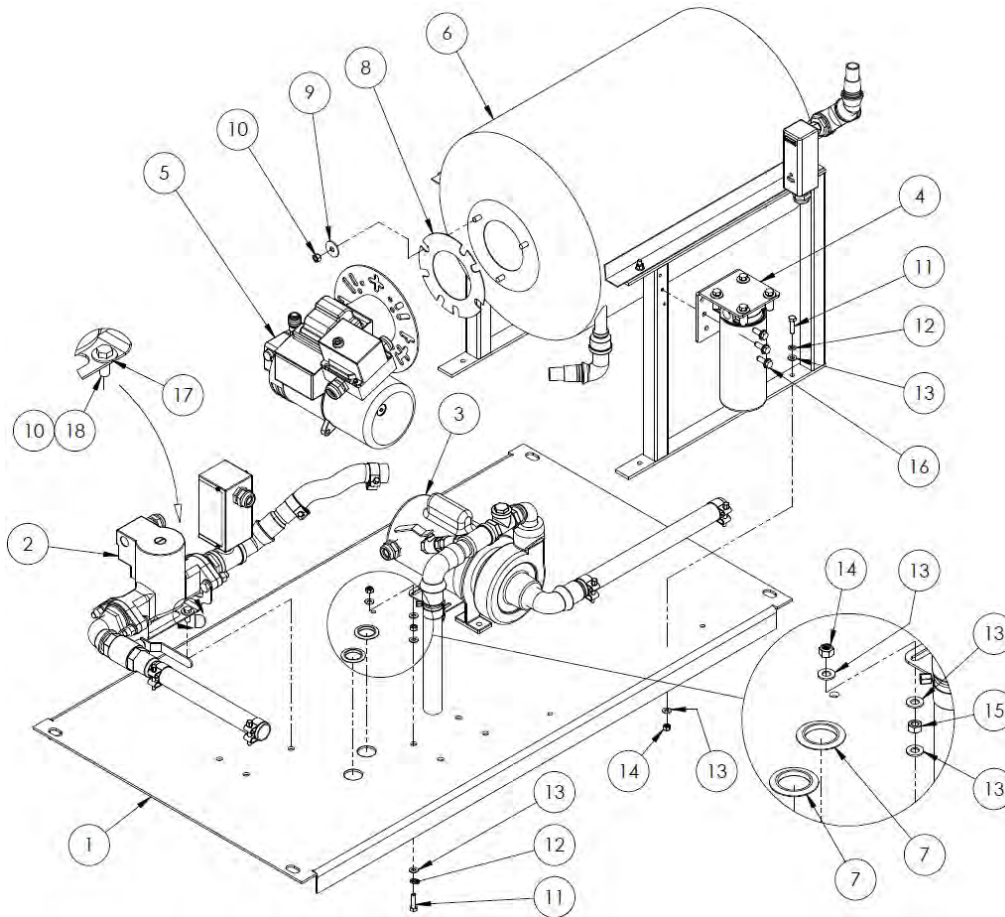
Ref.	Part	Description	Qty.
1	448156	REAR DOOR, RIGHT	1
2	448009	KEEPER, DOOR , 6 IN	1
3	448011	FLAP, 8 IN, REAR DOOR	1
4	448178	SIDE DOOR HINGE ASSY.	2
5	447160	SEAL, BULB	1

Side Door Assembly, HK 150



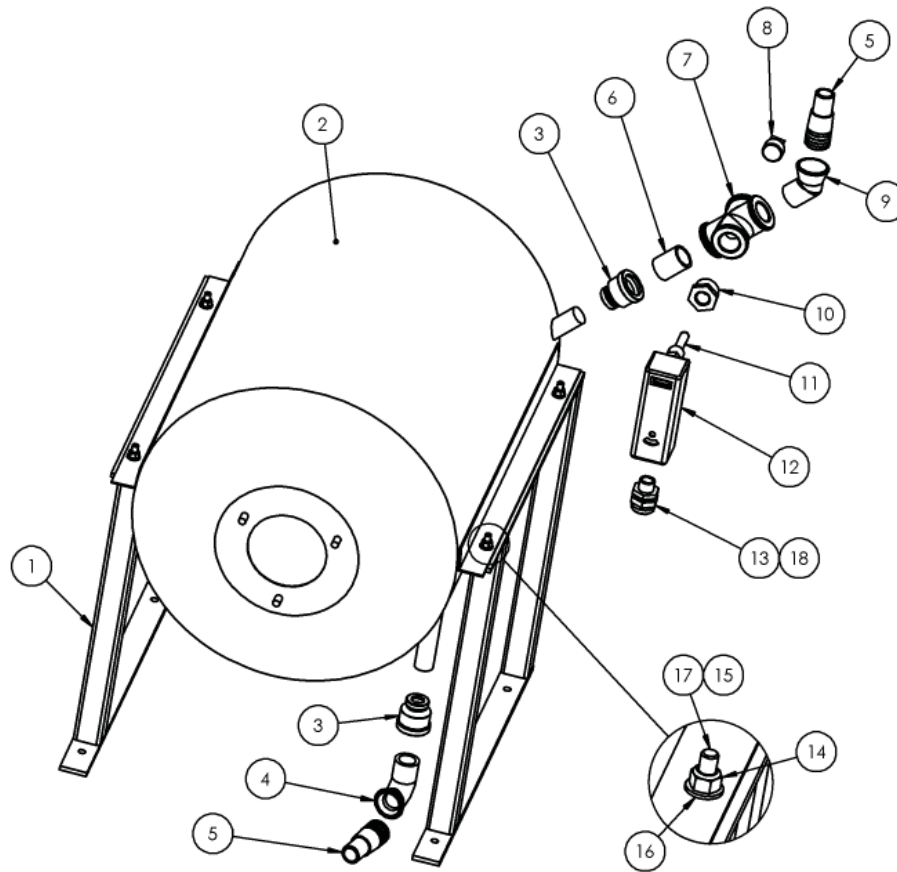
Ref.	Part	Description	Qty.
-	448157	COMPLETE SIDE DOOR ASSY	1
1	448136	PANEL, SIDE DOOR	1
2	448142	GUARD WIND SHIELD	1
3	448007	LATCH, 3 POINT, CENTER CASE	1
4	448006	HANDLE, DOOR	1
5	448178	SIDE DOOR HINGE ASSY.	2
6	448009	KEEPER, DOOR , 6 IN	1
7	447160	SEAL, BULB	1

Pump Plate Assembly, HK 150



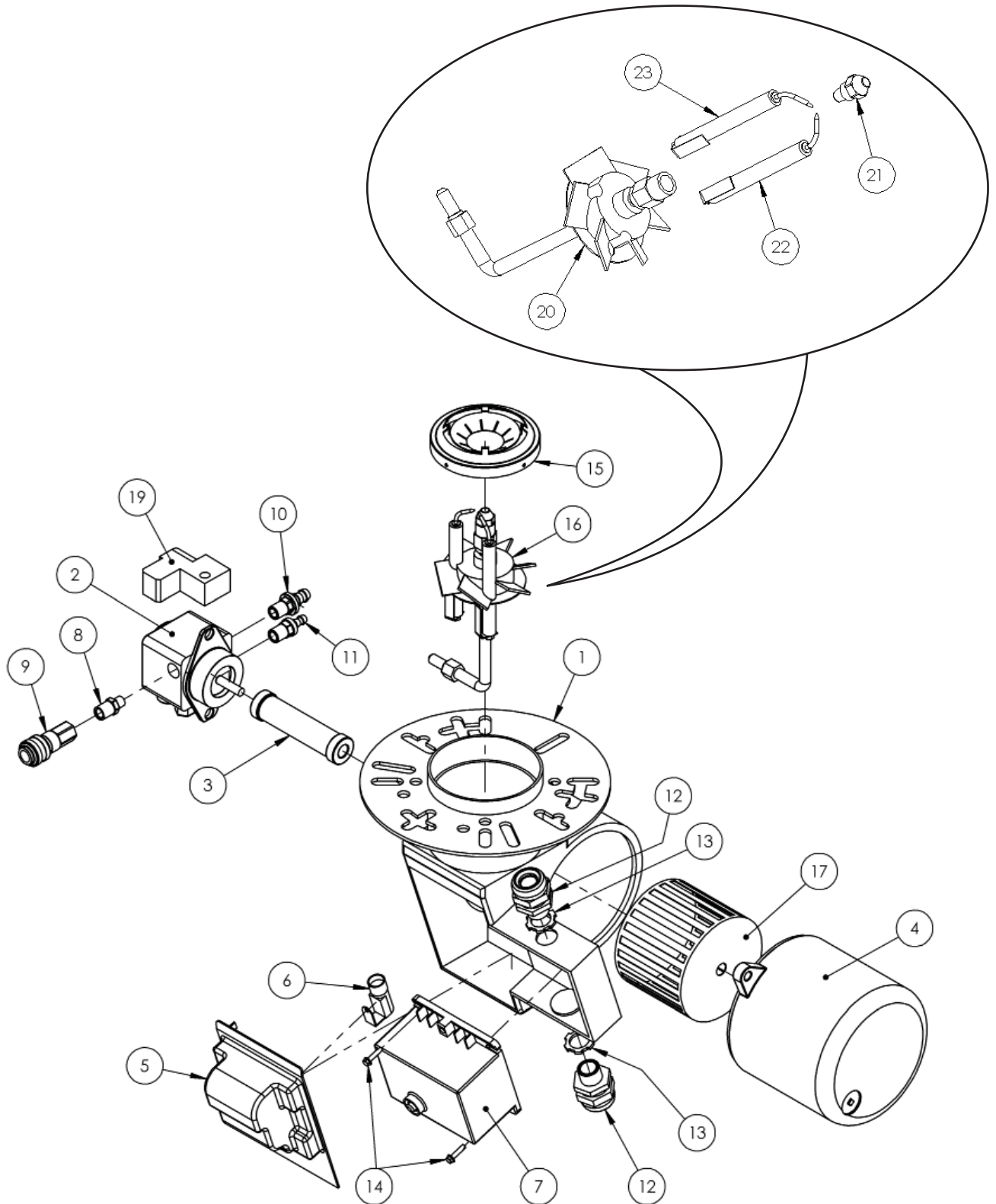
Ref.	Part	Description	Qty.
1	448103	PLATE, PUMP FLOOR, HK 150	1
2	See Page B-5	CIRCULATION PUMP ASSEMBLY, HK 150	1
3	See Page B-6	FIELD PUMP ASSEMBLY, HK 150	1
4	See Page B-8	FUEL FILTER ASSEMBLY, HK 150	1
5	451321	BURNER ASSEMBLY (See Page B-3)	1
6	See Page B-2	HEATING COIL ASSEMBLY, HK 150	1
7	440203	BUSHING, SNAP-IN	2
8	441136	SEAL, GASKET	1
9	174987	WASHER, 3/8 FLAT, WIDE	3
10	227830	NUT, HEX, 3/8	5
11	225990	BOLT, HEX HEAD, 5/16 X 1-1/4	12
12	176220	WASHER, 5/16, SPRING	12
13	174990	WASHER, FLAT, 5/16	40
14	227831	NUT, HEX, NYLON-INSERT, 5/16	12
15	227830	NUT, HEX, 3/8	8
16	228061	BOLT, FLANGE HEAD, SELF-DRILLING, 3/8 X 1	3
17	174986	WASHER, 3/8 PL	2
18	226210	BOLT, HEX HD 3/8 UNC X 1 PL	2

Heating Coil Assembly, HK 150

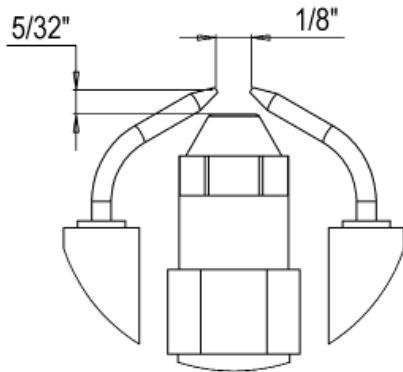


Ref.	Part	Description	Qty.
1	451060	STAND FRAME ASSEMBLY	2
2	451302	ASSEMBLY, COIL, HK 150	1
3	444816	BELL REDUCER, 1 x 1/2" GLVD	2
4	444820	ELBOW, 1" 90D STREET GLVD	1
5	444822	1" GALV, HOSE BARB	2
6	444740	1" GALVANIZED CLOSE NIPPLE	1
7	444850	CROSS, 1 x 1 x 1 GLVD	1
8	444866	PLUG, 1" NPT GLVD	1
9	446827	STREET ELBOW 1" 45 DEGREE GLVD	1
10	444817	REDUCER BUSHING, 1" x 1/2" GLVD	1
11	446543	HIGH LIMIT AQUASTAT WELL	1
12	446542	AQUASTAT, HIGH LIMIT HONELYWELL	1
13	446260	CONNECTOR, 1/2" STRESS RELIEF	1
14	225990	BOLT, HEX HEAD, 5/16 x 1-1/4	8
15	174990	WASHER, 5/16	4
16	176220	WASHER, 5/16, SPRING	4
17	227831	NUT, HEX, NYLON-INSERT 5/16	4
18	112510	NUT, LOCK CONDUIT 1/2	1

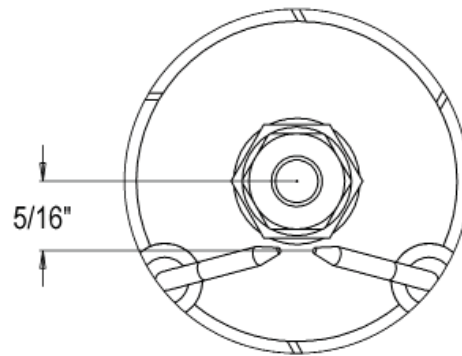
Burner Assembly, HK 150



Burner Assembly, HK 150 - Cont'd



PARTIAL SIDE VIEW

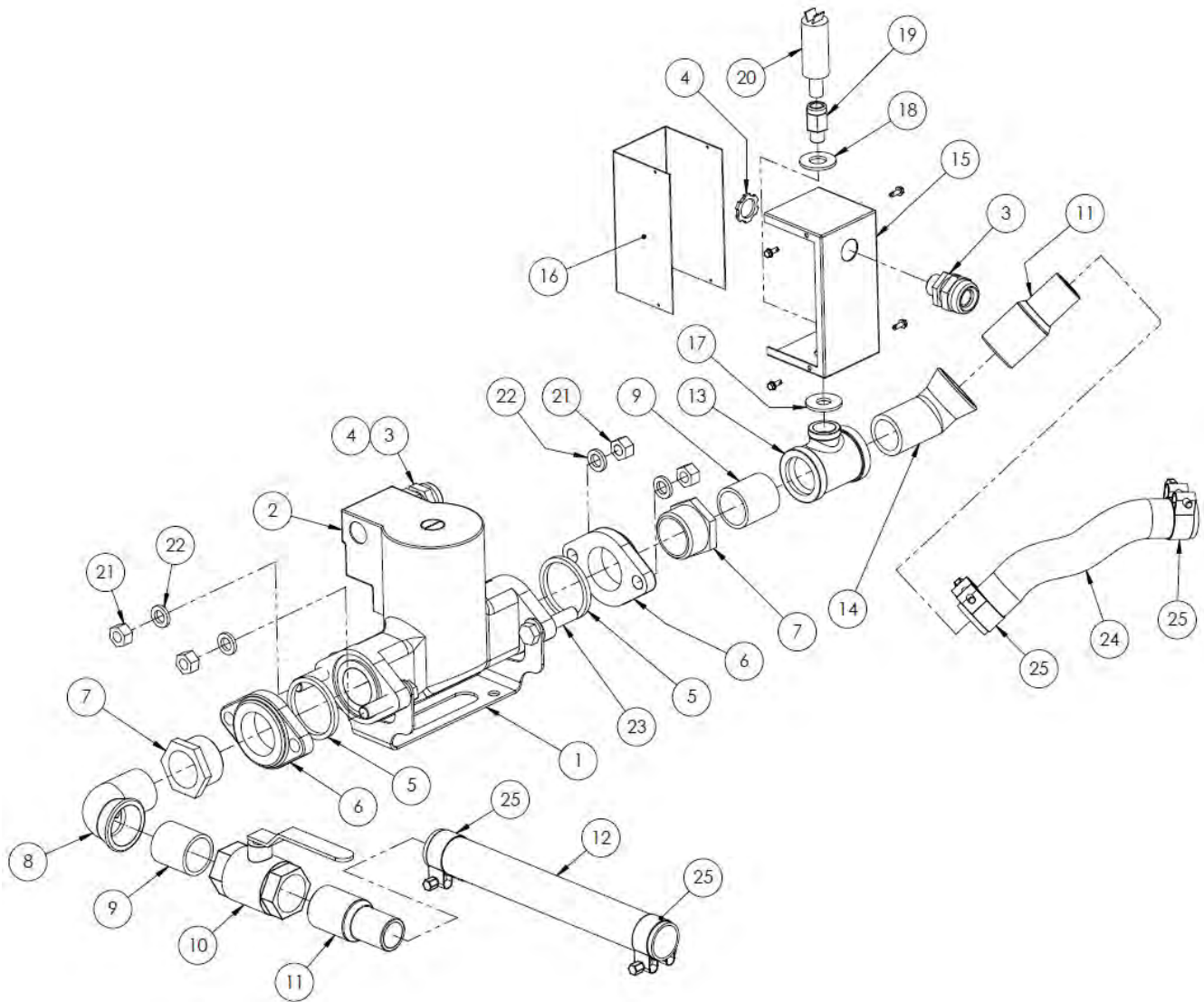


PARTIAL TOP VIEW

SETUP AT INSTALLATION OF ITEM 22 AND 23

Ref.	Part	Description	Qty.
1	449048	BURNER CHASSIS	1
2	449001	BURNER, FUEL PUMP 3450 RPM, 3 GPH	1
3	449032	BURNER, FLEXIBLE COUPLING	1
4	449024	BURNER, FAN MOTOR	1
5	449014	TRANSFORMER, IGNITOR	1
6	446491	SENSOR, CAD CELL LEAD	1
7	446490	SENSOR, PRIMARY CONTROL RELAY	1
8	444814	ADAPTER, 1/8 NPT x 1/4" MALE	1
9	444032	DISCONNECT, 1/4" FEMALE QUICK	1
10	444021	NIPPLE, 1/4 NPT x 3/8" BRSS HOSE	1
11	444042	NIPPLE, 1/4 NPT x 1/4" BRSS HOSE	1
12	446260	CONNECTOR, 1/2" STRESS RELIEF	2
13	112510	NUT, LOCK CONDUIT 1/2	2
14	162830	8-18 X 3/4 HWH SCREW	2
15	446766	BURNER, RETENTION HEAD	1
16	444615	BURNER, DRAWER ASSEMBLY - HK 150	1
17	449028	BURNER, FAN SQUIRREL CAGE	1
18	446768	BURNER, AIR BAND	1
19	446794	SOLENOID, FUEL	1
20	446615	BURNER DRAWER, ALKOTA	1
21	449053	NOZZLE, FUEL, 1.0 GPH 80D A	1
21*	449216	NOZZLE, FUEL, 0.8 GPH 80D A (High Altitude)	1
22	449003	BURNER, ELECTRODE	1
23	449030	BURNER, ELECTRODE	1

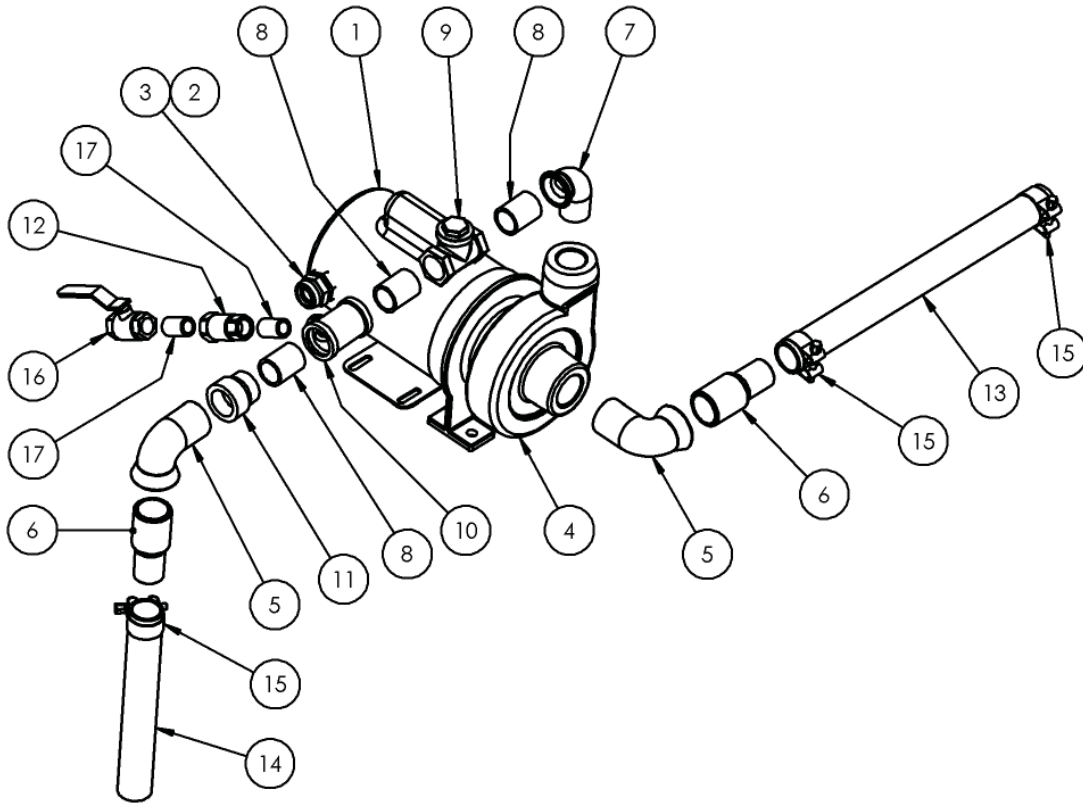
Circulation Pump, HK 150



Circulation Pump, HK 150 - Cont'd

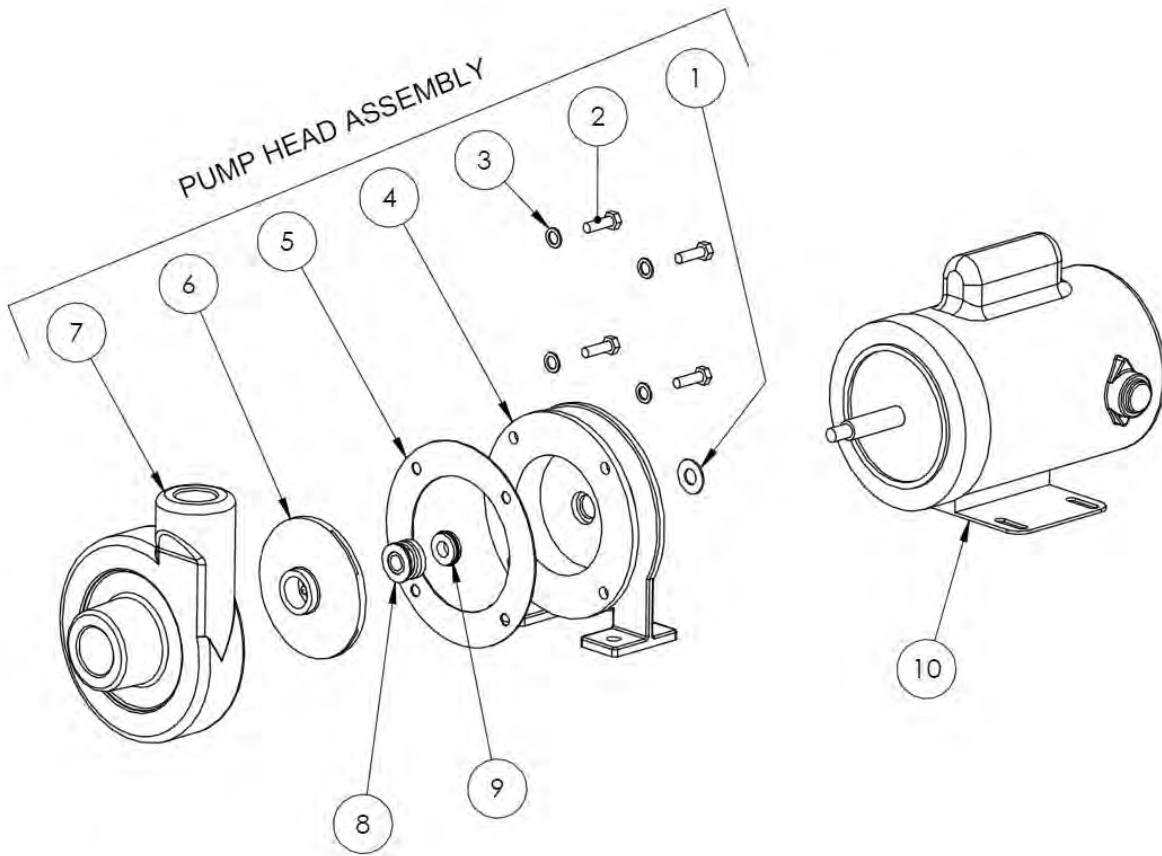
Ref.	Part	Description	Qty.
1	451101	BRACKET, MOUNTING PUMP	1
2	451306	PUMP, CIRCULATION	1
3	446260	CONNECTOR, 1/2" STRESS RELIEF	2
4	112510	NUT, LOCK CONDUIT 1/2"	2
5	446762	SEAL, CIRCULATING PUMP	2
6	451313	KIT, CLANGE - CIRCULATION	2
7	444749	BUSHING, 1-1/2 x 1-1/4 GLVD	2
8	444250	ELBOW, STREET 1-1/4 x 90D GLVD	1
9	444480	CLOSED NIPPLE, 1-1/4 GLVD	2
10	444110	BALL VALVE, 1-1/4 GLVD	1
11	444330	HOSE, BARB, 1-1/4 GLVD	2
12	442156	HOSE, TANK TO CIRCULATING PUMP	1
13	444540	TEE, REDUCER 1-1/4x1, 1-1/4x1/2	1
14	444750	ELBOW, 1-1/4 x 45D, STREET GLVD	1
15	446553 Bottom	CASE, HAMMOND - BOTTOM	1
16	446553 Top	CASE, HAMMOND - COVER	1
17	446613	WASHER, SMALL - PRESSURE SWITCH	1
18	446614	WASHER, LARGE - PRESSURE SWITCH	1
19	446523	SWITCH, PRESSURE SNUBBER	1
20	447011	SWITCH, PRESSURE	1
21	227854	NUT, HEX G2 PL 1/2 UNC	4
22	176250	WASHER, SPRING PL 3/8	4
23	226680	BPLT, HEX HD PL 1/2UNC x 2-1/2 LG.	4
24	442156	HOSE, COIL TO CIRCULATING PUMP	1
25	44810	T-BOLT, 1" CLAMP	4

Field Pump, HK 150



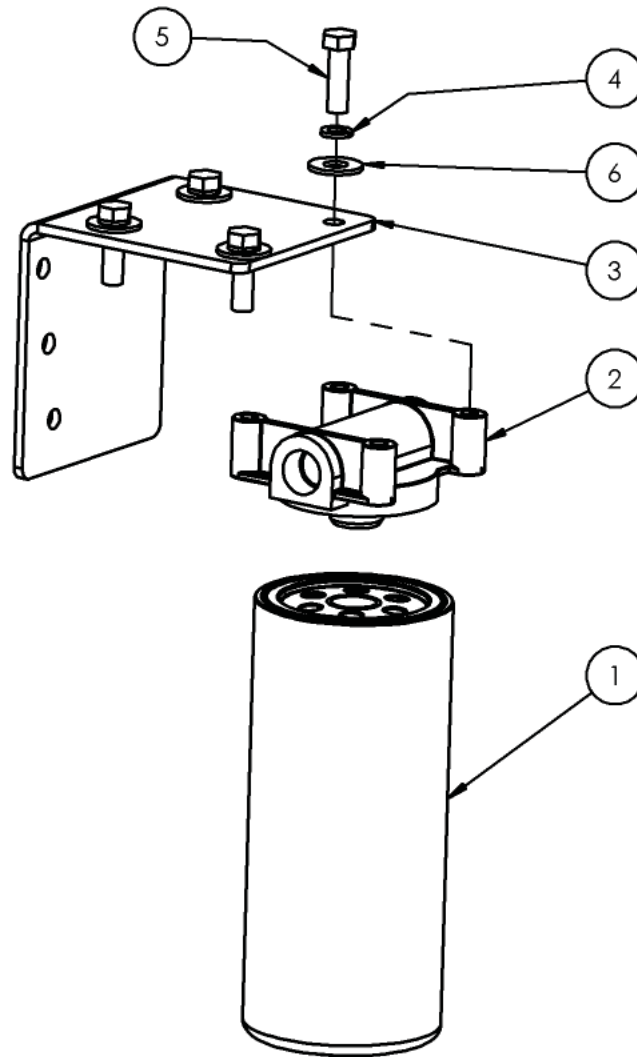
Ref.	Part	Description	Qty.
1	219110	MOTOR, 1/2 HP 115/230V	1
2	446260	CONNECTOR, 1/2" STRESS RELIEF	1
3	112510	NUT, LOCK CONDUIT 1/2	1
4	448045	PUMPHEAD, 1/2 HP (See Page B-7)	1
5	444250	ELBOW, STREET 1 1/4" 90D GLVD	2
6	444330	HOSE BARB, 1-1/4" GLVD	2
7	444820	ELBOW, 1" 90D STREET GLVD	1
8	444740	1" GALVANIZED CLOSE NIPPLE	3
9	444811	1" BRASS SWING CHECK VALVE	1
10	444821	1" x 1" x 1/2" GALVANIZED TEE	1
11	446800	BUSHING, REDUCER 1-1/4 x 1	1
12	444731	VALVE, 1/2" BRASS IN-LINE CHECK	1
13	442162	HOSE, 1-1/4", SUCTION	1
14	446815	HOSE, 1-1/4", 150 PSI	1
15	444810	T-BOLT, 1" CLAMP	3
16	444120	VALVE, BALL 1/2"	1
17	444440	NIPPLE, 1/2" CLOSE GLVD	2

Pump Assembly Breakdown, HK150



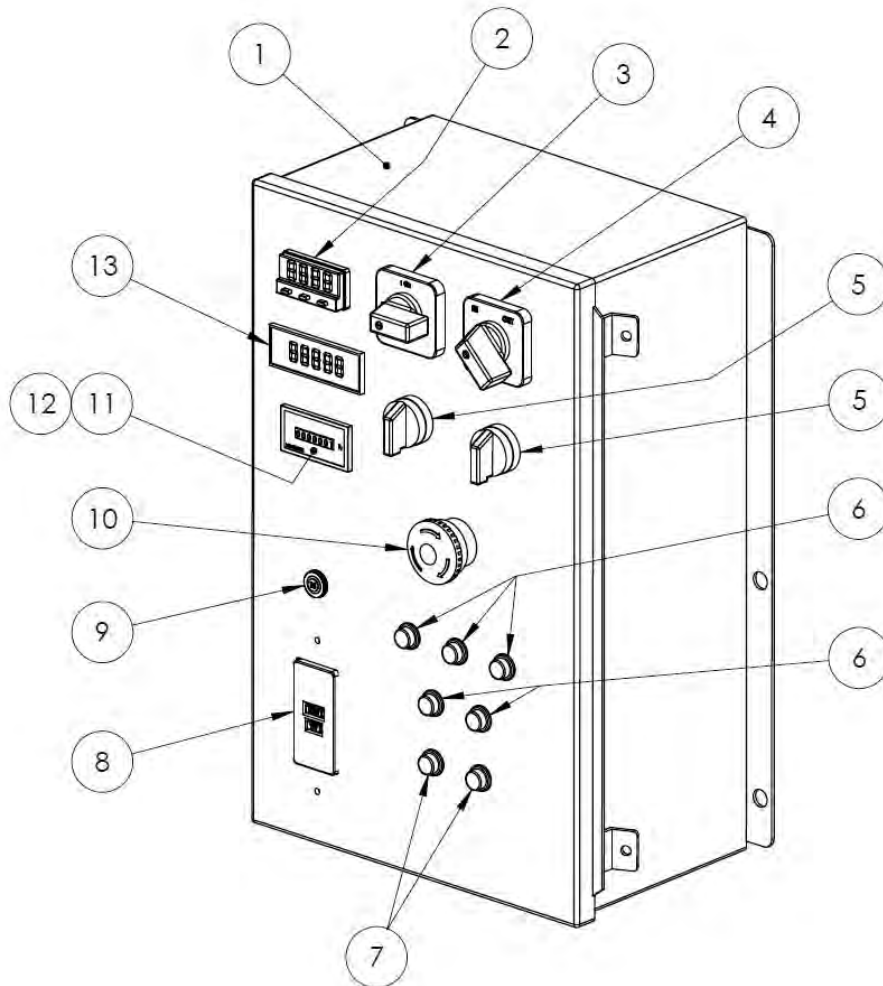
Ref.	Part	Description	Qty.
--	448045 **	PUMP HEAD ASSEMBLY, FAC-50-K	1
1	449002	FINGER SEAL	1
2	225980	BOLT, 5/16 UNC	4
3	175500	WASHER, 5/16	4
4	---	PUMP BODY, MOTOR SIDE	1
5	449012	GASKET, PUMP	1
6	---	PUMP Impeller	1
7	---	PUMP BODY, CAVITY SIDE	1
8	---	SEAL, MOTOR SIDE	1
9	---	SPRING SEAL	1
10	219110	MOTOR, 1/2HP 1-115/230V	1
	449059	SEAL KIT (Includes items: 8, and 9)	
	**	PUMP HEAD ASSEMBLY (Includes items: 1 to 7)	

Filter Assembly, HK150



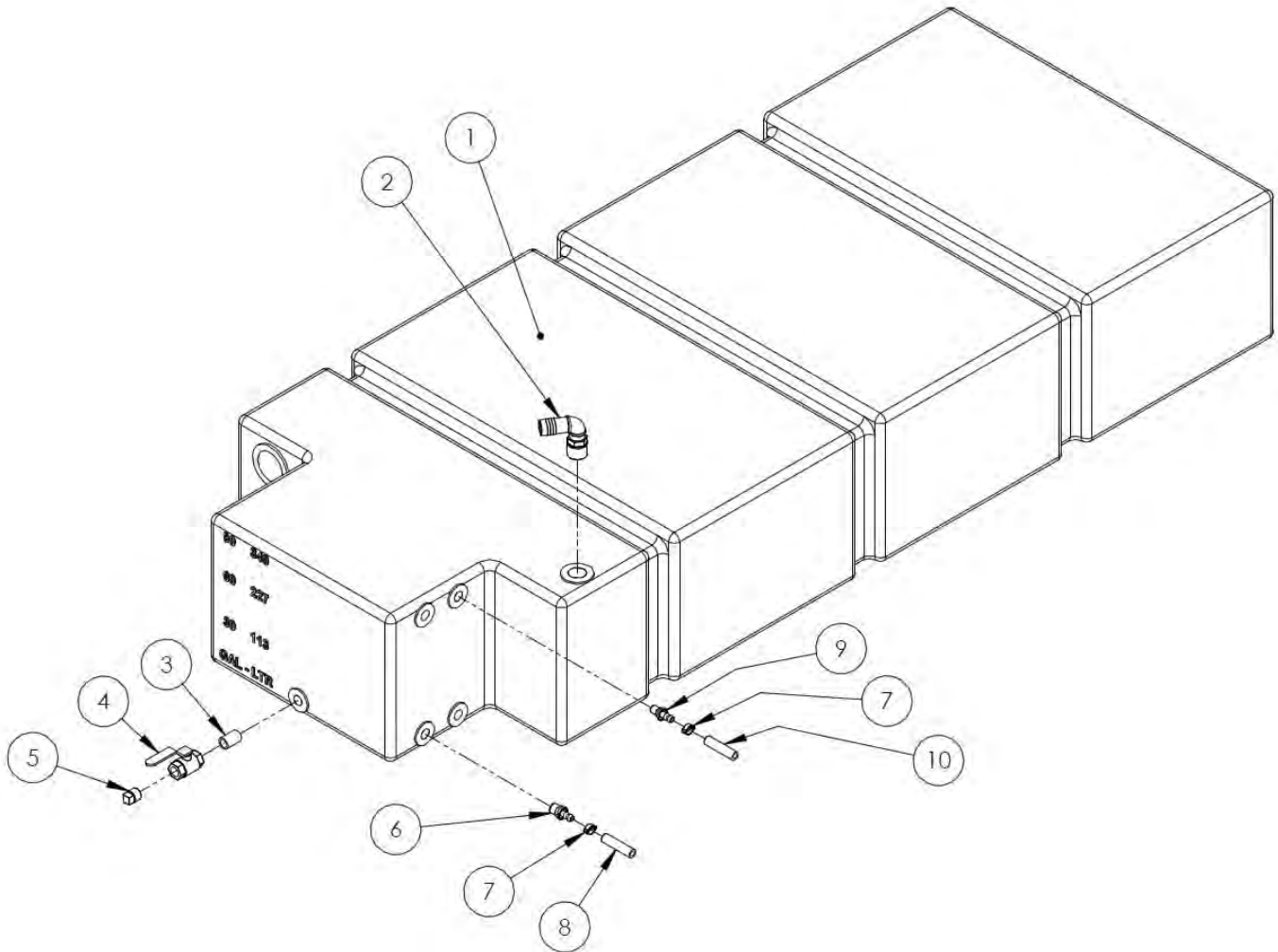
Ref.	Part	Description	Qty.
1	445157	FUEL FILTER	1
2	445156	FUEL FILTER HEAD	1
3	446821	BRACKET, FUEL FILTER	1
4	176220	WASHER, 5/16, SPRING	4
5	226220	BOLT, HEX HEAD 3/8 UNC	4
6	174986	WASHER, 3/8	4

Control Box Assembly, HK 150



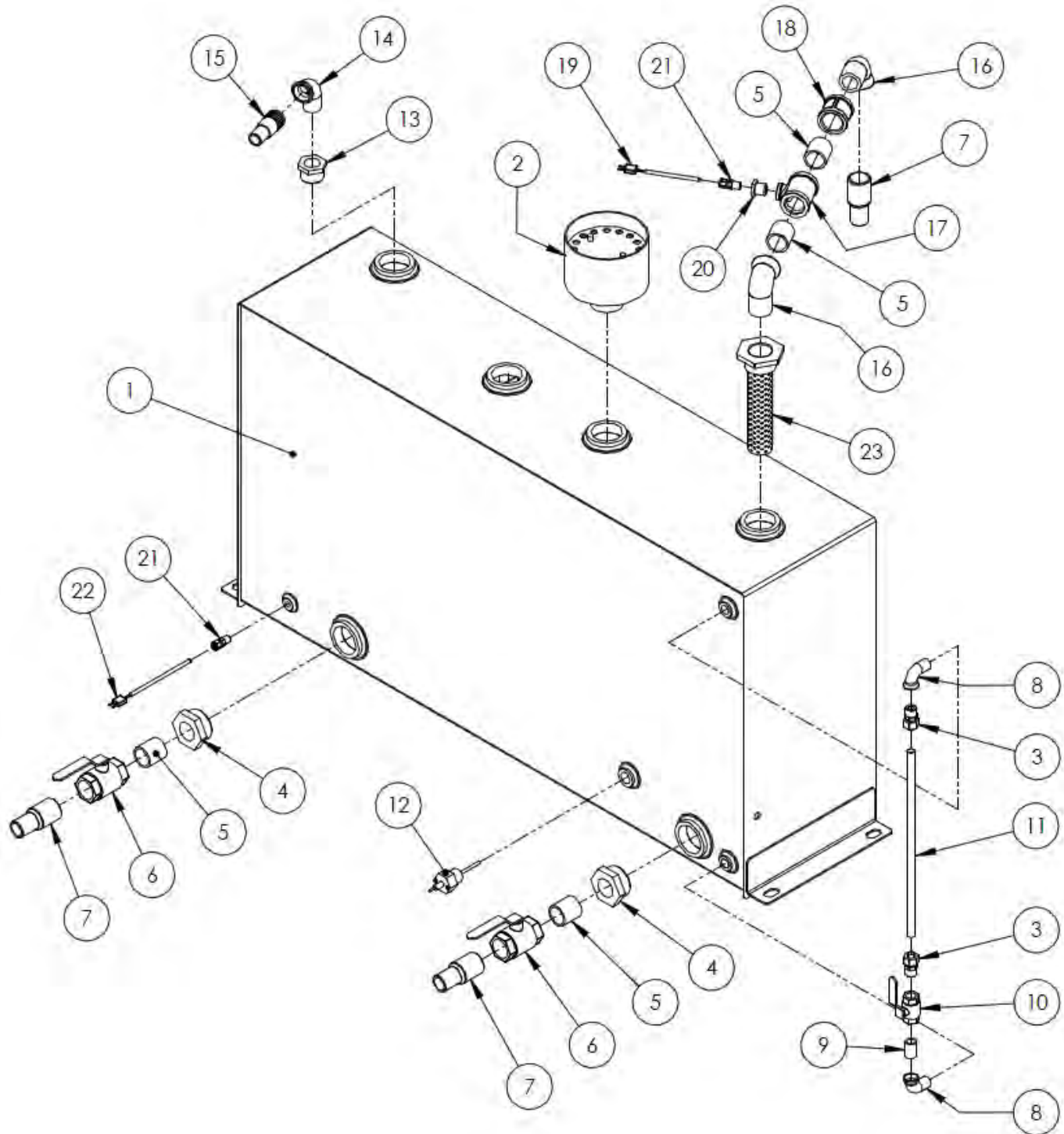
Ref.	Part	Description	Qty.
1	451350	BOX, ELECTRICAL	1
2	450093	TEMPERATURE CONTROLLER	1
3	446513	SWITCH, CIRCUIT POWER	1
4	446514	SWITCH, REEL DIRECTION	1
5	446517	SWITCH	2
6	446130	LIGHT, INDICATOR, AMBER	5
7	446140	LIGHT, INDICATOR, RED	2
8	446110	GROUND FAULT	1
9	446010	CIRCUIT BREAKER, 15 AMP	1
10	446527	SWITCH, EMERGENCY STOP	1
11	446120	HOUR METER	1
12	446125	HOUR METER, FACE PLATE	1
13	446519	RETURN READOUT	1

Fuel Tank Assembly, HK 150



Ref.	Part	Description	Qty.
1	451150	FUEL TANK, DIESEL	1
2	444829	ELBOW, 1" NPT X 1" 90 DEG. PVC	1
3	444440	NIPPLE, 1/2 CLOSE, GLVD	1
4	444120	VALVE, BALL 1/2"	1
5	444083	PLUG, 1/2" NPT, GLVD	1
6	444022	NIPPLE, 3/8 NPT x 3/8", LOL BRSS	1
7	444825	CLAMP, 3/8"	2
8	449203	HOSE, FUEL SUPPLY (10 FT. LENGTH)	1
9	444021	NIPPLE, 1/4 NPT x 3/8", BRSS HOSE	1
10	449204	HOSE, FUEL SYSTEM (10 FT. LENGTH)	1

Glycol Tank Assembly, HK 150

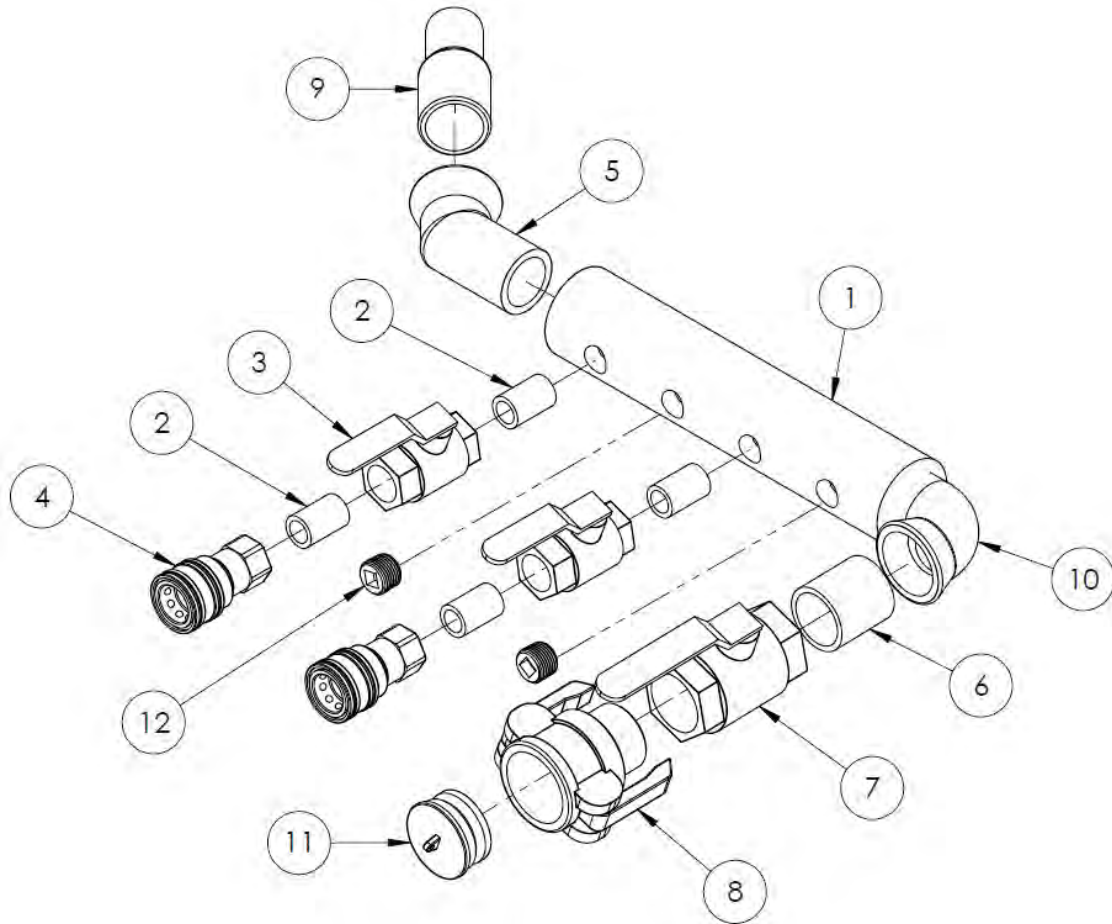


Glycol Tank Assembly, HK 150 - Cont'd

Ref.	Part	Description	Qty.
1	451050	GLYCOL TANK WELDMENT	1
2	444704	BREATHER, 500MBTU ASSY	1
3	444200	COMPRESSION, 1/2NPT X 5/8"	2
4	444140	BUSHING, 2 X 1 1/4" GLVD	2
5	444480	CLOSE NIPPLE, 1-1/4, GLVD	4
6	444110	BALL VALVE, 1-1/4	2
7	444330	HOSE BARB, 1-1/4, GLVD	3
8	444610	ELBOW, STREET 1/2" GLVD	2
9	444440	NIPPLE, 1/2 CLOSE, GLVD	1
10	444120	VALVE, BALL 1/2"	1
11	446756	POLY TUBING, 1/2ID X 5/8"OD	1
12	441140	LOW LEVEL CUT OFF - PROBE	1
13	444818	REDUCER, 1 1/4x1" BUSHING GLVD	1
14	444820	ELBOW, 1" 90D STREET GLVD	1
15	444822	1" GALV, HOSE BARB	1
16	444250	ELBOW, STREET 1-1/4 X 90DEG, GLVD	2
17	444540	TEE, REDUCER 1 1/4X1 1/4X1/2"	1
18	444807	COUPLING, 1 1/4" GLVD	1
19*	446518	RETURN THERMOCOUPLE ASSY	1
20	444832	BUSHING, 1/2"x1/4" GLVD	1
21	444758	FITTING, COMPRESSION	2
22*	446704	GLYCOL TANK THERMOCOUPLE ASSY.	1
23	445846	RETURN STRAINER	1

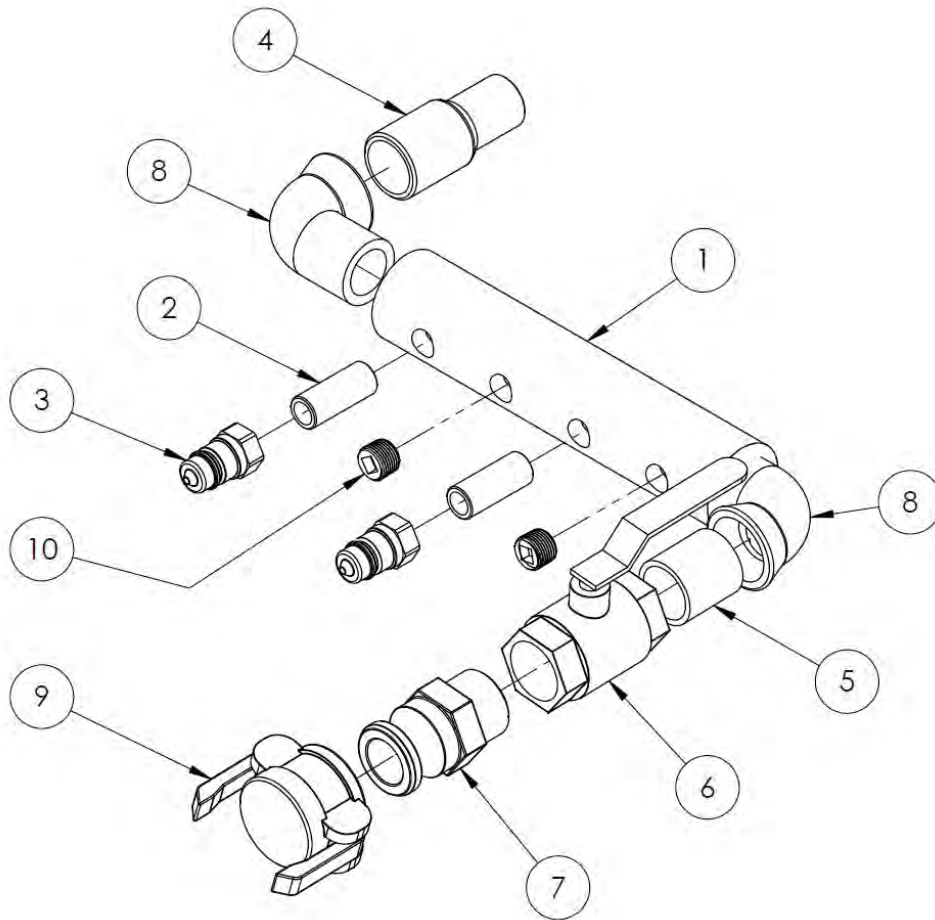
* Order this item with Compression Fitting - Item 21

Supply Manifold, HK 150



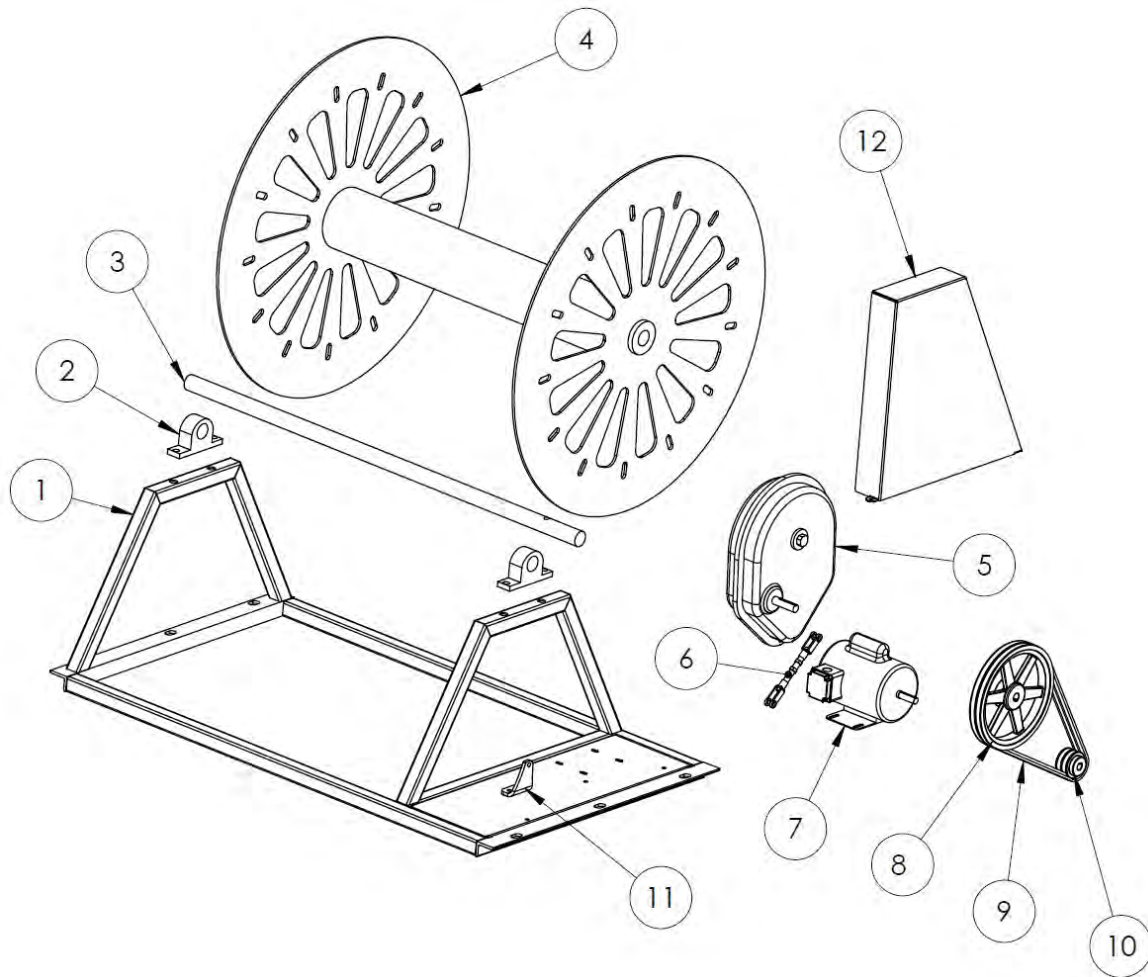
Ref.	Part	Description	Qty.
1	451200	MANIFOLD, 4 PLACE	1
2	444440	NIPPLE, 1/2 CLOSE, GLVD	8
3	444081	VALVE, BALL 3/4 NPT, BRASS	4
4	444901	QUICK CONNECT, 1/2, FEMALE	4
5	444750	ELBOW, 1-1/4 X 45 DEG STREET, GLVD	1
6	444480	CLOSE NIPPLE, 1-1/4, GLVD	1
7	444110	VALVE, BALL 1-1/4	1
8	444751	CAMLOCK, 1-1/4 NPT MAL X 1-1/4 FEM	1
9	444330	HOSE BARB, 1-1/4, GLVD	1
10	444250	ELBOW, STREET 1-1/4 X 90 DEG, GLVD	1
11	444086	PLUG, 1-1/4, CAMLOCK, MALE	1
12	444083	PIPE PLUG, 1/2 NPT	2

Return Manifold, HK 150

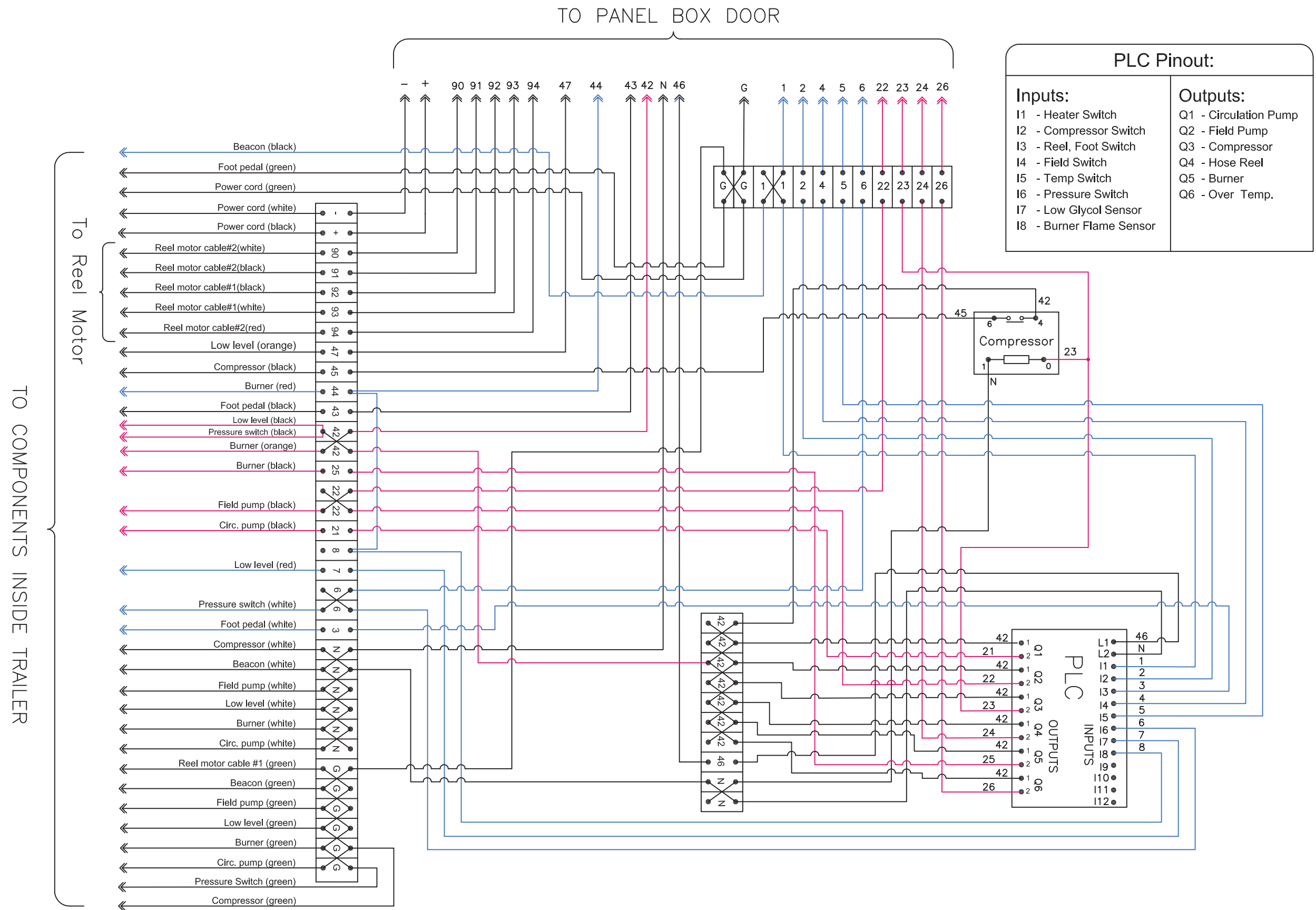


Ref.	Part	Description	Qty.
1	451200	MANIFOLD, 4 PLACE HK150	1
2	444805	NIPPLE, 1/2 X 2, GLVD	4
3	444900	QUICK CONNECT, 1/2, MALE	4
4	444330	HOSE BARB, 1-1/4, GLVD	1
5	444480	CLOSE NIPPLE, 1-1/4, GLVD	1
6	444110	BALL VALVE, 1-1/4	1
7	444754	CAMLOCK, 1-1/4NPT X 1-1/4 MALE	1
8	444250	ELBOW, STREET 1-1/4 X 90DEG, GLVD	2
9	444087	PLUG, 1-1/4, CAMLOCK, FEM	1
10	444083	PIPE PLUG, 1/2 NPT	2

Reel Assembly, HK 150

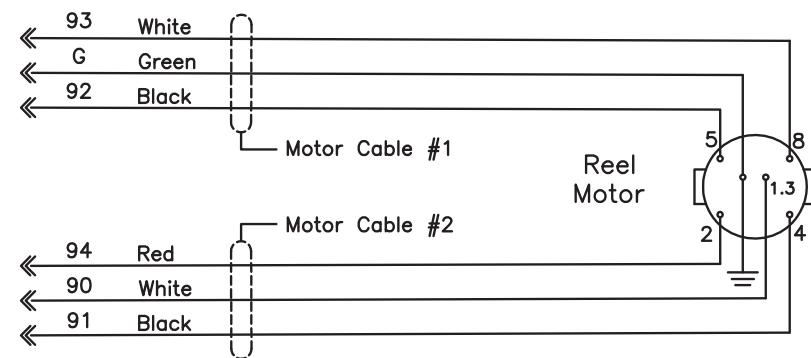


Ref.	Part	Description	Qty.
1	448107	REEL FRAME ASSEMBLY	1
2	451309	PILLOW BLOCK	2
2*	451310	REPLACEMENT PILLOW BLOCK INSERT	2
3	451082	REEL SHAFT	1
4	451078	HOSE REEL SPOOL	2
5	496645	GEAR REDUCER	1
6	448025	TURNBUCKLE ASSEMBLY	1
7	219210	ELECTRIC MOTOR	1
8	496160	PULLEY, LARGE	1
9	239591	V BELT	2
10	430950	PULLEY, SMALL	1
11	451088	REEL TURNBUCKLE BRACKET	1
12	451118	BELT GUARD	1



HK150 PANEL BOX ELECTRICAL SCHEMATIC

TO TERMINAL STRIP IN PANEL BOX

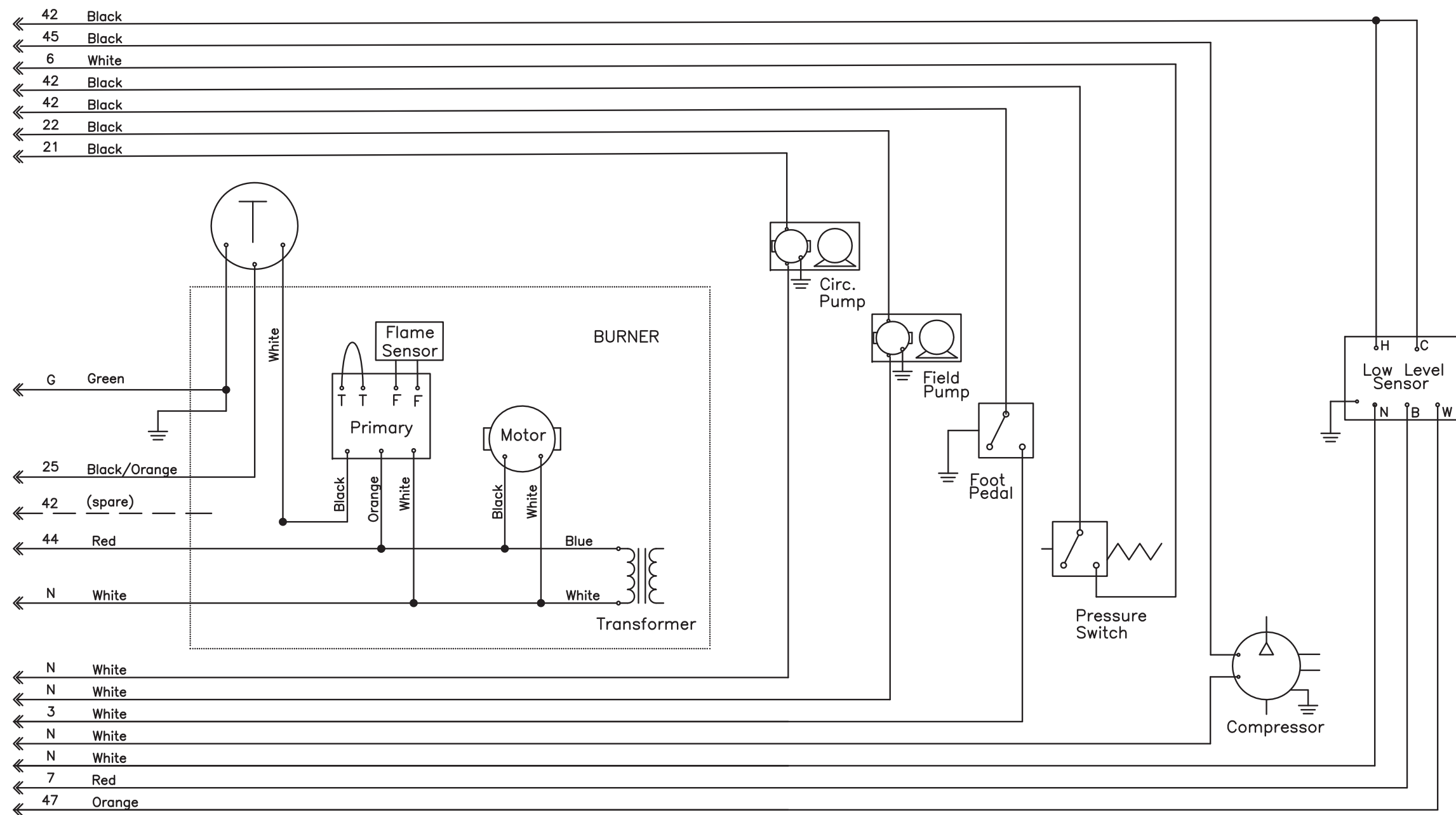


Voltage on reel direction switch while foot pedal is pressed	
In Direction:	Out Direction:
L1 - L2 = 120V	L1 - L2 = 120V
L1 - 1.3 = 120V	L2 - 1.3 = 120V
L2 - 8 = 120V	L2 - 5 = 120V

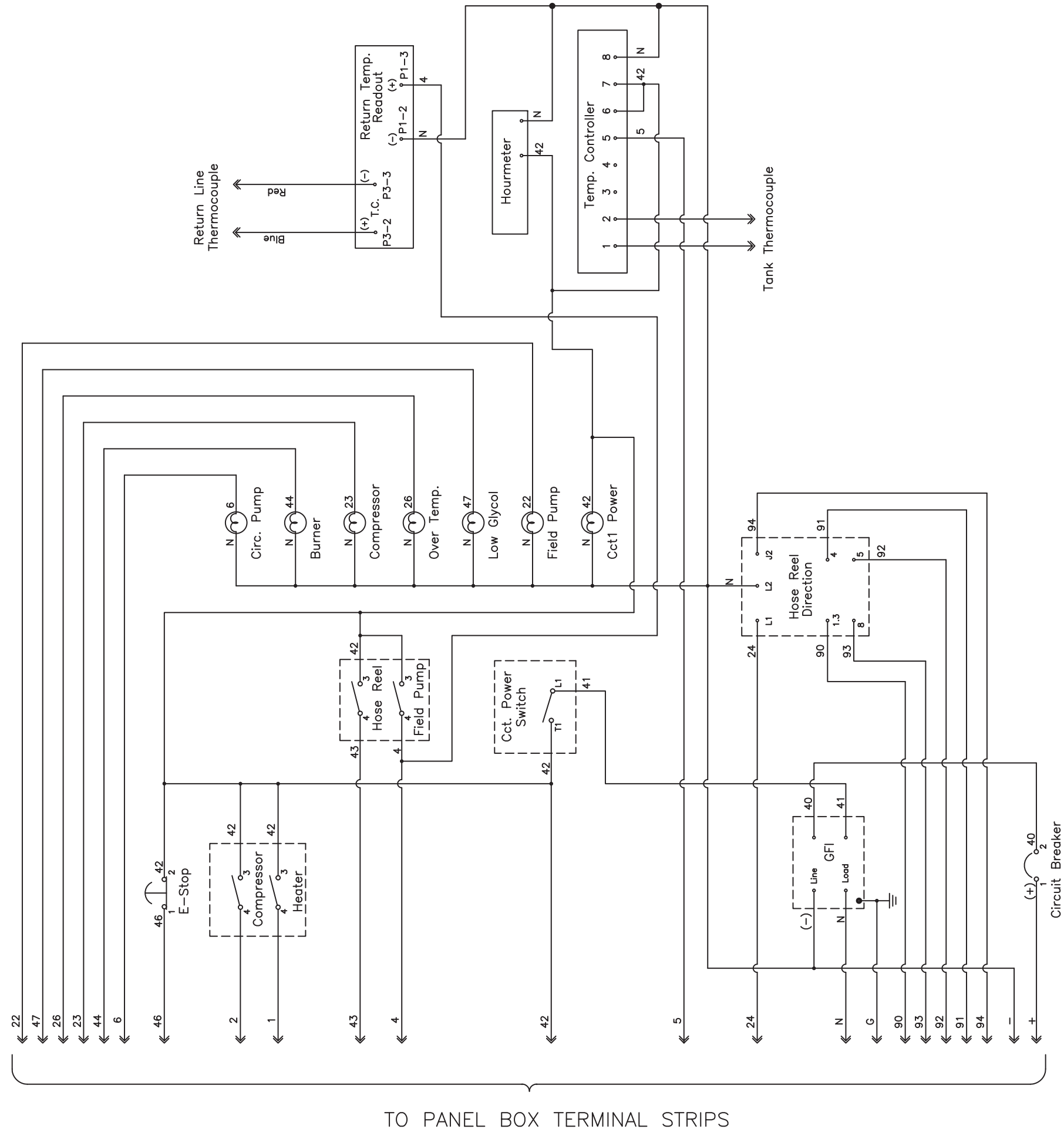
No voltage when pedal not pressed



ALL GROUNDS ARE CONNECTED IN PANEL BOX (GREEN)



HK150 INSIDE TRAILER ELECTRICAL SCHEMATIC



TO PANEL BOX TERMINAL STRIPS

HK150 PANEL DOOR ELECTRICAL SCHEMATIC



Product Brochure



Contact Us

HEAT KING

Toll Free: 1-800-661-0304
Phone: 204-885-5800
Fax: 204-885-7557

P.O. Box 234 Station L
Winnipeg, Manitoba R3H 0Z5
Canada

